Mohit Kori

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LinkedIn | **GitHub** | **Portfolio**

Professional Summary

Embedded Systems & Electronics Engineer with dual MSc degrees in Power Electronics & Control and Electronics, combined with hands-on experience in embedded firmware development, hardware validation, industrial electronics testing, and automation workflows. Skilled in STM32-based embedded systems, Embedded C/C++, MATLAB/Simulink, PCB design, PLC programming, and real-time debugging. Experienced in root cause analysis, hardware/software troubleshooting, and test validation using oscilloscopes, multimeters, and simulation tools. Strong background in technical problem-solving, process optimisation, and cross-functional engineering support within industrial and high-throughput environments.

Technical Skills

- **Embedded Systems & Firmware:** Embedded C/C++, STM32, ARM Cortex-M, ADC, PWM, SPI, I2C Timers, Interrupts, UART, GPIO, Sensor Interfacing, Real-Time Embedded Systems
- **AI & Software Fundamentals:** Python, Machine Learning Fundamentals, Embedded AI Interest, TinyML Concepts, Data Analysis, Algorithmic Problem Solving
- **Automation & Control Systems:** PID Control, PLC Fundamentals, Siemens TIA Portal, Industrial Automation Concepts, Industrial Sensors, Process Control
- **Electronics & Hardware:** PCB Design, Signal Conditioning, Power Electronics, DC/DC Converters, Circuit Debugging, Hardware Validation, EMI/EMC Awareness
- **Testing & Diagnostics:** Oscilloscope, Multimeter, Functional Testing, Root Cause Analysis (RCA), Troubleshooting, Calibration, HIL Testing
- **Software & Engineering Tools:** MATLAB/Simulink, LTSpice, VS Code, KiCad, EasyEDA, MS Excel (Pivot Tables, VLOOKUP)
- **Engineering Practices:** Technical Documentation, Cross-Functional Collaboration, Structured Problem Solving, Prototype Development, System Validation, Continuous Improvement

Technical Projects

STM32-Based Line-Following & Obstacle-Aware System (GitHub)

- Developed embedded firmware in Embedded C/C++ on STM32 ARM Cortex-M platform for real-time motor and sensor control
- Implemented ADC-based sensor acquisition, interrupt-driven logic, and PWM motor control reducing response delay by ~18%
- Applied PID control algorithm improving tracking stability and reducing oscillation during navigation
- Performed hardware/software debugging using oscilloscopes, GDB-style debugging, and structured root cause analysis
- Validated system behaviour through iterative testing and firmware optimization
  Technologies: STM32, Embedded C/C++, PWM, ADC, PID, ARM Cortex-M, MATLAB

Autonomous Hazard Inspection Robot (GitHub)

- Developed autonomous embedded navigation system using sensors and real-time obstacle detection logic
- Built MATLAB/Simulink models for system-level validation and hardware-in-loop testing
- Improved operational stability through iterative tuning and debugging workflows
- Conducted simulation-to-hardware verification using real embedded hardware
  Technologies: Embedded C, MATLAB/Simulink, Sensors, PWM, HIL Testing

Custom PCB Design (KiCad & EasyEDA)

- Designed schematic and PCB layout for microcontroller-based embedded system using KiCad and EasyEDA
- Applied signal routing and power integrity principles during PCB development
- Conducted DRC validation and pre-prototyping verification improving design reliability
- Supported hardware debugging and embedded system integration workflows
  Technologies: KiCad, EasyEDA, PCB Design, Hardware Validation

Professional Experience

Operations Supervisor **Nov 2022 – Apr 2026**
Amazon DLU2, Luton, United Kingdom

- Coordinated daily high-throughput operations handling 5,000+ units/day across cross-functional teams
- Led operational troubleshooting and root cause analysis improving process metrics from 96% to 98%
- Reduced operational delays by ~15% through workflow optimisation and escalation management
- Managed communication between operations, logistics, and delivery stakeholders ensuring service continuity
- Supported data-driven decision-making using Excel reporting and operational performance tracking

Electronics & Electrical Test Engineer **Jan 2021 – Sep 2022**
Multispan Control Instruments Pvt. Ltd., Ahmedabad, India

- Tested and troubleshot 50+ industrial electronic systems including embedded power supply and DC/DC converter modules
- Improved fault detection efficiency by ~20% through structured debugging and validation procedures
- Diagnosed PCB-level electrical faults using oscilloscopes, multimeters, and hardware analysis techniques
- Reduced recurring technical failures by ~20% using systematic root cause analysis methodologies
- Generated 30+ technical reports and compliance documentation supporting traceability and quality assurance

Academic Qualifications

- **MSc in Power Electronics and Control** **Sep 2022 – Oct 2023**
  University of Hertfordshire, United Kingdom – Graduated with Commendation
- **MSc in Electronics** **Jun 2018 – Apr 2020**
  Sardar Patel University, Gujarat, India – Graduated with Distinction
- **BSc in Instrumentation** **Jun 2015 – Apr 2018**
- Sardar Patel University, Gujarat, India – Graduated with Distinction

Additional Information

- **Certification:** Siemens TIA Portal PLC Programming
- **Languages:** English (Fluent), German (A2 – improving)
- **Availability:** Immediate Joining Available
- **Work Authorization:** Germany Opportunity Card (Eligible for full-time roles)